FR-07 – Step 4: Boundary value analysis

Input: the classes in `FR-07_2_equivalence_classes.md` and the test cases in `FR-07_3_test_cases.md`.
Method: lecture S04, step 4.

Method

- BVA applies to **ordered** domains (quantities, counts, prices, IDs, displayed numbers). For each ordered class, test **LB−1, LB, LB+1, UB−1, UB, UB+1**, plus the smallest/largest value the UI allows (slide 26, points 8 *and 9*).
- A boundary point already hit by a step 3 test case is **reused, not duplicated**.
- Valid boundary points may share a test case; each invalid boundary point gets its own.
- **M** = the largest quantity accepted, found by FR-07_TC_24. **N** = the number of products in the catalog (count them on the home page after seeding).
- Test data and setups P1–P4 are the same as in step 3 (A = iPhone 15 Pro Max, 30.000.000 ₫).

Variables with no ordered domain (no BVA)

Variable	Why no BVA
IN_03 Add-to-cart source, IN_07 Delete, IN_08 Dialog answer, IN_09 Continue shopping	Buttons and choices: sets, not ordered.
IN_11 Login state, IN_12 Navigation type	Sets.
IN_14 `name` via API	Text with no length rule; the server must ignore it (D7).
IN_17 Auth token	Set {valid, missing, tampered}.

Boundary table

V = expected accepted, I = expected rejected.

B1. Quantity on the detail page (IN_02) – valid range [1, M]

Point	Value	V / I	Test case
Smallest via UI (8*)	`-99999999999`	I	**FR-07_BVA_02**
LB−1	`0`	I	FR-07_TC_22
LB	`1`	V	**FR-07_BVA_01**
LB+1	`2`	V	FR-07_TC_01
UB−1	M − 1	V	**FR-07_BVA_03**
UB	M	V	**FR-07_BVA_04**
UB+1	M + 1 (first rejected value)	I	FR-07_TC_24
Largest via UI (9*)	up to `1e21`	I	FR-07_TC_24

B2. Quantity typed in the cart (IN_18) – valid range [1, M]

Point	Value	V / I	Test case
LB−1	`0`	I	FR-07_TC_28
LB	`1`	V	**FR-07_BVA_05**
LB+1	`2`	V	**FR-07_BVA_05**
UB−1	M − 1	V	**FR-07_BVA_06**
UB	M	V	**FR-07_BVA_06**
UB+1	M + 1	I	FR-07_TC_30

B3. + and − buttons (IN_05, IN_06) – result must stay in [1, M]

Point	Action	V / I	Test case
"+" at LB	1 → 2	V	**FR-07_BVA_07**
"−" at LB+1	2 → 1	V	**FR-07_BVA_07**
"−" at LB	at 1 (must be disabled)	I	FR-07_TC_26
"+" at UB−1	M − 1 → M	V	**FR-07_BVA_08**
"+" at UB	at M	I	FR-07_TC_25

B4. `quantity` in `POST /api/cart` (IN_15) – valid range [1, M]

Point	Value	V / I	Test case
LB−1	`0`	I	FR-07_TC_40
LB	`1`	V	**FR-07_BVA_09**
LB+1	`2`	V	FR-07_TC_15
UB−1	M − 1	V	**FR-07_BVA_10**
UB	M	V	**FR-07_BVA_10**
UB+1	M + 1	I	FR-07_TC_42

B5. Number of distinct products in the cart (IN_10) – range [0, N]

Point	Value	V / I	Test case
LB	0 (empty state)	V	FR-07_TC_07
LB+1	1	V	FR-07_TC_01
LB+2	2	V	FR-07_TC_03
UB−1	N − 1	V	**FR-07_BVA_11**
UB (9*)	N (every product in the catalog)	V	**FR-07_BVA_11**

UB+1 is impossible: there is no (N+1)-th product to add.

B6. Size of the displayed totals (OUT_04, OUT_05) – output must stay exact and in `đ` format

Line totals and the cart total are outputs computed from price × quantity. Very large results are where systems lose precision (digits rounded) or switch format (for example to `3e+18`). Expected

values are computed with exact arithmetic (e.g. Python or Windows Calculator), never with a spreadsheet, which rounds after 15 digits.

Point	Quantity of A	Exact expected line total	Test case
Quantity = M	M	M × 30.000.000 ₫	**FR-07_BVA_04**
The D2 starting probe	`99999999999`	2.999.999.999.970.000.000 ₫	**FR-07_BVA_12**
Just below 10^{21}	`33333333333333`	999.999.999.999.990.000.000 ₫	**FR-07_BVA_13**
Just above 10^{21}	`33333333333334`	1.000.000.000.000.020.000.000 ₫	**FR-07_BVA_13**

BVA_12 and BVA_13 only apply if M is at least that large; otherwise they are "Not executed".

B7. Cart badge number (OUT_14) – display width

The badge shows total units (D4). The number of digits is an ordered output: a badge built for one or two digits can be cut off or replaced by "99+".

Point	Total units	Test case
0	0	FR-07_TC_07
1 digit, upper	9	**FR-07_BVA_14**
2 digits, lower	10	**FR-07_BVA_14**
2 digits, upper	99	**FR-07_BVA_14**
3 digits, lower	100	**FR-07_BVA_14**

B8. `price` in `POST /api/cart` (IN_13) – only the catalog price (30.000.000) is accepted as-is

D7: whatever price the client sends, the cart must use the catalog price. Values just around the catalog price test whether the server compares exactly or only roughly.

Point	Value	V / I	Test case
Lowest non-negative	`0`	I	**FR-07_BVA_15**
Catalog − 1	`29999999`	I	**FR-07_BVA_16**
Catalog	`30000000`	V	FR-07_TC_15
Catalog + 1	`30000001`	I	**FR-07_BVA_17**

B9. `id` in `POST /api/cart` (IN_16) – valid range [1, N]

Point	Value	V / I	Test case
LB−1	`0`	I	**FR-07_BVA_18**
LB	`1`	V	FR-07_TC_15
UB	N	V	**FR-07_BVA_19**
UB+1	N + 1	I	**FR-07_BVA_20**

Assumes IDs run 1 … N without gaps after seeding; check with the detail-page URLs.

New BVA test cases

Cleanup after each: empty the cart. "Exact" totals: compare every digit, not just the first few.

TC ID	Test case name	Boundary	Precondition	Test steps (with data)	Expected result
FR-07_BVA_01	Detail page quantity 1	B1 LB	P1, P2	A's detail page, quantity `1`, "Add to cart".	A × 1 in the cart; total 30.000.000 ₫; badge 1.
FR-07_BVA_02	Detail page, very large negative	B1 8*	P1, P2	A's detail page, quantity `-99999999999` , "Add to cart".	Invalid-quantity error; cart unchanged; no negative total.
FR-07_BVA_03	Detail page quantity M − 1	B1 UB−1	P1, P2, P4	Quantity M − 1, "Add to cart".	Accepted; line total exactly (M − 1) × 30.000.000 ₫.
FR-07_BVA_04	Detail page quantity M	B1 UB; B6	P1, P2, P4	Quantity M, "Add to cart".	Accepted; line total and total exactly M × 30.000.000 ₫, in `₫` format with separators.
FR-07_BVA_05	Cart: type 1, then 2	B2 LB, LB+1	P1, P2; cart has A × 5	1. Type `1`, confirm. 2. Type `2`, confirm.	1. Quantity 1; total 30.000.000 ₫. 2. Quantity 2; total 60.000.000 ₫.
FR-07_BVA_06	Cart: type M − 1, then M	B2 UB−1, UB	P1, P2, P4; cart has A × 2	1. Type M − 1, confirm. 2. Type M, confirm.	Both accepted; totals exact.
FR-07_BVA_07	"+" from 1, then "−" back to 1	B3 "+" at LB, "−" at LB+1	P1, P2; cart has A × 1	1. Click "+". 2. Click "−".	1. Quantity 2; "−" enabled. 2. Quantity 1; "−" now disabled (D3).
FR-07_BVA_08	"+" from M − 1 to M	B3 "+" at UB−1	P1, P2, P4; cart has A × (M − 1)	Click "+".	Quantity M; total exact; "+" now disabled or blocked.
FR-07_BVA_09	API quantity 1	B4 LB	P1, API	`POST /api/cart` with `"quantity":1`.	HTTP 200; cart has A × 1.

TC ID	Test case name	Boundary	Precondition	Test steps (with data)	Expected result
FR-07_BVA_10	API quantity M − 1 and M	B4 UB−1, UB	P1, API, P4	1. `POST /api/cart` with `"quantity": M − 1`; check, then empty the cart. 2. Same with M.	Both accepted; `GET /api/cart` shows the exact quantity each time.
FR-07_BVA_11	Every product in the cart	B5 UB−1, UB	P1, P2	1. Add every product except one (quantity 1 each); check the cart. 2. Add the last one; check again.	N − 1 rows, then N rows; total = sum of all catalog prices; badge N; page still readable.
FR-07_BVA_12	Total at quantity 99.999.999.999	B6	P1, P2; M ≥ 99999999999	A's detail page, quantity `99999999999`, "Add to cart"; open the Cart.	Line total and total exactly **2.999.999.999.970.000.000 ₫**: every digit correct, no exponent.
FR-07_BVA_13	Totals around 10^{21}	B6	P1, P2; M ≥ 33333333333334	1. Quantity `33333333333333`; check the cart; empty it. 2. Quantity `33333333333334`; check the cart.	1. Exactly **999.999.999.999.990.000.000 ₫**. 2. Exactly **1.000.000.000.000.020.000.000 ₫**, shown as digits (not `1e+21`).

TC ID	Test case name	Boundary	Precondition	Test steps (with data)	Expected result
FR-07_BVA_14	Badge with 9, 10, 99, 100 units	B7	P1, P2; cart has A × 1	Type `9`, then `10`, `99`, `100` in A's quantity field, checking the badge after each.	Badge shows 9, 10, 99, 100 exactly; not cut off and not "99+".
FR-07_BVA_15	API price 0	B8	P1, API	`POST /api/cart` with `"price":0`.	Cart uses 30.000.000 ₫ (D7); total is not 0.
FR-07_BVA_16	API price just below catalog	B8	P1, API	`POST /api/cart` with `"price":29999999`.	Cart uses 30.000.000 ₫.
FR-07_BVA_17	API price just above catalog	B8	P1, API	`POST /api/cart` with `"price":30000001`.	Cart uses 30.000.000 ₫.
FR-07_BVA_18	API product ID 0	B9 LB−1	P1, API	`POST /api/cart` with `"id":0`.	Rejected; cart unchanged.
FR-07_BVA_19	API product ID N	B9 UB	P1, API	`POST /api/cart` with `"id":` N and that product's name and price.	HTTP 200; cart has product N.
FR-07_BVA_20	API product ID N + 1	B9 UB+1	P1, API	`POST /api/cart` with `"id":` N + 1.	Rejected; cart unchanged.

Summary

	Count
Ordered domains analysed	9 (B1–B9)

	Count
Boundary points identified	47
Points already hit by step 3 test cases (reused)	17
Points needing new test cases	30
New BVA test cases (several valid points share one)	20
Total FR-07 test cases (step 3 + step 4)	**48 + 20 = 68**

Notes for review

1. **Seven test cases depend on M** (BVA_03, 04, 06, 08, 10, 12, 13). If TC_24 finds no limit, record "no upper limit" as the finding and mark those that cannot be run as "Not executed".
2. **B6 and B7 are output boundaries:** they check numbers the system displays, not numbers the user types. BVA_12 starts from D2's probe value, so it should run even if M turns out to be larger.
3. **Use exact arithmetic for expected totals.** Excel and Google Sheets round after 15 significant digits, so they would give wrong expected values for BVA_12 and BVA_13.
4. **B8 is an interpretation of D7.** The spec only says nothing about client prices; the tester decided the server must always use the catalog price, so any difference, even ±1 ₫, is a bug under that decision.
5. **B9 assumes IDs 1 … N without gaps.** If the detail-page URLs show gaps, use the lowest and highest real IDs instead.